Python -DB Connectivity

Objective

- Connection
- Methods
- MySQL Connection
- Mongo db Connection
- Summary

Connection

- For communicating with oracle database through our Python program we require cx_Oracle module.
- **Installation**
  - **pip install cx-Oracle**

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Methods

OBJECT™ TECHNOLOGIES

- connect(): Establishes a connection between the Python program and Oracle database.

```
con = cx_Oracle.connect('username/password@localhost')
```

- cursor(): To execute a SQL query and to provide results.

```
cursor = cx_Oracle.cursor()
```

- execute/executemany method :
  - cursor.execute(sqlquery) -> to execute a single query.
  - cursor.executemany(sqlqueries) -> to execute a single query with multiple bind variables/place holders.
  - commit(): commit() the result reflects in the database for DML(Data Manipulation Language) queries that comprise operations like update, insert, delete.

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- fetchone(), fetchmany(int), fetchall():
  - fetchone() : This method is used to fetch one single row from the top of the result set.
  - fetchmany(int): This method is used to fetch a limited number of rows based on the argument passed in it.
  - fetchall() : This method is used to fetch all rows from the result set.
  - close(): After all done it is mandatory to close all operations.
    - cursor.close()
    - con.close()

MySQL Connection

Installation

pip install mysql-connector

Steps to connect a python application to MySQL Database

1. Import mysql.connector module
2. Create the connection object.
3. Create the cursor object
4. Execute the query

Create a database named "mydatabase":

```
import mysql.connector
mydb = mysql.connector.connect(
  host="localhost",
  user="yourusername",
  password="yourpassword"
)
mycursor = mydb.cursor()
mycursor.execute("CREATE DATABASE mydatabase")
```


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Mongo db Connection

1. Install Driver

$ pip install pymongo

2. Create Python File

```
from pymongo import MongoClient # import mongo client to connect
import pprint
# Creating instance of mongoclient
client = MongoClient()
# Creating database
db = client.javatpoint
employee = {"id": "101", “name": "Shivraj","profession": "Software
Engineer“}
# Creating document
employees = db.employees
# Inserting data
employees.insert_one(employee)
# Fetching data
pprint.pprint(employees.find_one())
```


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Summary

OBJECT™ TECHNOLOGIES

- Connection
- Accessing Data
- Manipulating Data
- Other Queries
- MySQL Connection
- Mongo db Connection

Thank You